INCH-POUND
MIL-F-85421B
29 August 1994
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29 May 1986

MILITARY SPECIFICATION

FITTINGS, TUBE, FLUID SYSTEMS, SEPARABLE, BEAM SEAL, 3000/4000 PSI, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design requirements and test procedures for separable beam seal fittings which includes end fittings (see 6.10.3), fitting bodies (see 6.10.6), and boss fittings (see 6.10.1) for use in aerospace fluid systems. Design requirements are for class 3000 in corrosion resistant steel (CRES) only and for class 4000 in titanium alloy (Ti) only. Definition of fittings and related terms are defined in 6.10.

1.2 Classification. Fittings shall be of the following type and pressure classes of the hydraulic system as specified below:

1.2.1 Type

a. Type II System - Temperature range -65°F to +275°F (-54°C to +135°C).

1.2.2 Pressure Classes

a. Class 3000 - 3000 psi (21,000 kPa) nominal operating pressure where the cutout pressure at the main pressure controlling device is 3000 psig ±15% (21,000 kPa ±15%).
b. Class 4000 - 4000 psi (28,000 kPa) nominal operating pressure where the cutout pressure at the main pressure controlling device is 4000 psig ±15% (28,000 kPa ±15%).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code SR3, Highway 547, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4730
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for
MIL-H-5606 - Hydraulic Fluid, Petroleum Base; Aircraft, Missile and Ordnance
MIL-S-8879 - Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for
MIL-B-43666 - Boxes, Shipping, Consolidation
MIL-L-46010 - Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting
MIL-H-83282 - Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft, Metric, NATO Code Number H-537

STANDARDS

MILITARY

MIL-STD-100 - Engineering Drawing Practices
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage (Part 1 of 4 Parts)
MIL-STD-280 - Definitions of Item Levels, Item Exchangeability, Models and Related Terms
MIL-STD-2073-1 - DoD Material Procedures for Development and Application of Packaging Requirements

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(See supplement 1 for list of associated specification sheets)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP, Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

UNITED STATES GOVERNMENT PRINTING OFFICE

GPO Style Manual

(Copies of the GPO Style Manual are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

ASME-B46.1 - Surface Texture

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D3951 - Packaging, Commercial
ASTM-D1974 - Fiberboard Shipping Containers, Methods of Closing, Sealing, and Reinforsing, Standard Practice for

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-AIR1377 - Fire Test Equipment for Flexible Hose and Tube Assemblies
SAE-AMS4921 - Titanium, Bars, Forgings, and Rings, Annealed, 70,000 psi (485 Mpa) Yield Strength (DoD Adopted)

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SAE-AMS4928 - Titanium Alloy, Bars, Forgings, & Rings 6AL-4V. Annealed, 120,000 PSI (825 MPA) Yield Strength (DoD Adopted)
SAE-AMS4945 - Titanium Alloy Tubing, Seamless, Hydraulic - 3.0Al - 2.5V, Texture Controlled, 105,000 PSI (724 MPA) Yield Strength
SAE-AMS4965 - Titanium Alloy, Bars, Forgings, & Rings, 6.0 Al -4.0V, Solution & Precipitation Heat Treated (DoD Adopted)
SAE-AMS5561 - Tubing, Steel, Welded & Drawn, Corrosion & Heat Resistant 9.0Mn - 20 Cr - 6.5Ni - 0.28N, High Pressure Hydraulic (DoD Adopted)
SAE-AMS5637 - Steel Bars, Corrosion Resistant 18 Cr - 9.0Ni (SAE 30302) Cold Drawn, 125,000 PSI (860 MPA), Tensile Strength
SAE-AMS5643 - Steel Bars, Forgings, Tubing and Rings, Corrosion Resistant, 16 Cr-4.0 Ni- 0.30 (Cb+Ta)-4.0 Cu Solution Heat Treated
SAE-AMS5656 - Bar, Steel, Forgings & Rings, Corrosion Resistant 9.0Mn - 20Cr - 6.5Ni - 0.27N (DoD Adopted)
SAE-AMS5659 - Bar, Forgings, and rings, and Extrusions Corrosion Resistant 15 Cr 0 45 Ni - 0.30 (CD+Ta) - 3.5 Cu Consumable Electrode Melted Solution Heat Treated, Precipitation Hardenable (DoD Adopted)
SAE-ARP603 - Hose, Hydraulic, Tubing & Fitting Assemblies, Impulse Testing of (DoD Adopted)
SAE-ARP1185 - Flexure Testing of Hydraulic Tubing Joints and Fittings
SAE-AS1055 - Hose, Flexible, Tube Assemblies, Coils, Fittings & System Components, Fire Testing of (DoD Adopted)

(Copies of SAE publications may be obtained from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15098.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of conflict between the requirements of this specification and the specification sheet, the specification sheet shall govern.

3.2 Qualification. The fittings and fitting bodies furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.3 and 6.6).

3.3 Material.

3.3.1 Fittings. The material used shall be titanium alloys and corrosion resistant steels as specified in Table I. The corrosion resistant steel fittings are designed for class 3000 system. The titanium fittings are designed for class 4000, but may be used in both class 3000 and class 4000 systems. The material shall be heat treated, as required, and have surface protection to meet the test requirements of this specification. Recycled materials shall be utilized to the maximum extent possible provided the materials used conform to the applicable material specifications.

TABLE I. Fitting and tubing materials.

Item Name	Titanium (Class 4000)	CRES (Class 3000)
Nuts Internally Threaded	SAE-AMS4965	SAE-AMS5643 or SAE-AMS5659 Condition H1075
Fitting Bodies	SAE-AMS4965, SAE-AMS4928, SAE-AMS4921	SAE-AMS5643 or SAE-AMS5659 Condition H1075 or SAE-AMS5656
Tubing	SAE-AMS4945	SAE-AMS5561
Swivel Wires	SAE-AMS5637 Protected with Solid Film Lubricant per MIL-L-46010 Type I	

3.3.2 Tubing. The tubing used for the fabrication of the test fitting assemblies shall be in accordance with Tables I and II.

3.4 Design and construction.

3.4.1 Configuration and dimensions. The configuration and dimensions shall conform to requirements of this document and the design requirements specified on the applicable specification sheets of MIL-F-85421/1 through MIL-F-85421/5 and MIL-F-85421/10 through MIL-F-85421/17 (see supplement 1).

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TABLE II. Fitting and tubing data and stress level for flexure test.

Tubing Size			3000 PSI (21,000 kPa) Operating Pressure CRES Tubing				4000 PSI (28,000 kPa) Operating Pressure Ti Tubing			
Dash No	OD		Wall Thickness		Stress		Wall Thickness		Stress	
	in	mm	in	mm	psi	kPa	in	mm	psi	kPa
-03	.188	4.76	.020	.508	24,000	168,000	.020	.508	20,000	140,000
-04	.250	6.35	.020	.508	24,000	168,000	.020	.508	19,000	133,000
-05	.313	7.94	.020	.508	24,000	168,000	.028	.711	18,000	126,000
-06	.375	9.53	.020	.508	22,000	154,000	.028	.711	18,000	126,000
-08	.500	12.30	.026	.660	20,000	140,000	.035	.889	17,000	119,000
-10	.625	15.88	.033	.838	18,000	126,000	.044	1.118	17,000	119,000
-12	.750	19.05	.039	.991	16,000	112,000	.052	1.320	17,000	119,000
-14	.875	22.22	-	-	-	-	.061	1.549	16,000	112,000
-16	1.000	25.40	.052	1.32	15,000	105,000	.070	1.778	13,000	91,000
-20**	1.250	31.75	.032*	.813	15,000	105,000	.088	2.235	13,000	91,000
-24**	1.500	38.10	-	-	-	-	.054*	1.372	4,000	28,000

LEGEND:

* 500 psig (3800 kPa) systems for suction lines.

** Sizes -20 and -24 were originally designated as -21 and -25 on revision A of this specification. These sizes will not connect with the original -20 and -24 fittings. See 6.5 for additional information.

Notes:

1. All test assembly wall thicknesses and lengths are shown in Table II and Table IV.
2. Stress values specified are the minimum bending stress for the flexture test per 4.6.6.
3. For test purposes, tubing diameters and wall thicknesses shall be per Table II. If different tube thicknesses are used for aircraft design, additional tests may be required. For use of tubing wall thickness less than .020 inch (.5 mm), special approval must be obtained from the contracting activity.

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3.4.2 Drill offset. On fittings where the fluid passage is drilled from each end, the offset between the drilled holes at the meeting point of the drills shall not exceed .015 inch (.4mm). Angular misalignment shall not exceed two degrees.

3.4.3 Reduction in fluid passage (bore) cross-section. Straight and nonreducer shape fittings shall be bored through out the passage such that a ball having a diameter, .020 inch (.5mm) less than the minimum specified bore shall pass through the passage. For nonstraight and nonreducer shape fittings (such as crosses, tees, and elbows etc,) the cross-sectional area at the junction of the fluid passage shall not be smaller than the cross-sectional area of the smallest passage.

3.4.4 Tube attachment. The method of joining the tubing to the end fitting (brazing, welding, mechanical attachment, swaging, etc.) shall not adversely affect the properties, strength, or geometry of tubing or end fitting. The joint shall be of a quality compatible with the design and performance requirements of this specification and shall permit inspection by direct measurement, x-ray, or other means.

3.4.5 Installation tooling. Size of tool envelope required to install fittings on tubing in aircraft shall be kept to a minimum. The wrench flats or pads of the fittings shall permit use of standard wrenches.

3.4.6 Fitting body design. Fitting body shall be designed to prevent failure (see 6.10.4) of sealing surfaces. The sealing interface shall be integral with the fittings. Removable seals are not permitted. Prior to assembly, the female nut shall be pull back in such position that the threaded end of the nut is flush within ±.0625 inch (±1.59 mm) of the outer beam surface. When assembled the fitting body to maximum torque (see 6.10.9) the gap between the threaded end of the nut and hex face of the male fitting body shall be a minimum of .030 inch (.8 mm).

3.4.7 Fitting body interface. All male and female end fittings shall be tested using fitting bodies with end configurations conforming to the design standards MIL-F-85421/1 and MIL-F-85421/2. All male fitting bodies tested shall be interchangeable with male fitting bodies dimensioned as shown in the design standards.

3.4.8 Threads. All threads shall be in accordance with MIL-S-8879. The surface finish of the thread flanks shall be 63 microinches Ra per ASME-B46.1. Internal threads shall be coated with solid film lubricant per MIL-L-46010, Type I.

3.5 Performance. The fitting assemblies (see 6.10.5) shall meet the performance requirements of 3.5.1 through 3.5.8.

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3.5.1 Proof pressure. The fitting assembly shall withstand pressure equal to twice the nominal operating pressure of the system for five minutes without leakage, evidence of permanent deformation, or other malfunction that shall affect assembly or disassembly using the torque values per Table III when tested in accordance with 4.6.2.

3.5.2 Burst pressure. The fitting assembly shall not rupture or leak (no more than allowable) at any pressure less than or equal to four times the nominal operating pressure. The test shall be conducted in accordance with 4.6.3 for a minimum of five minutes at burst pressure. The fitting assembly shall not have to meet any disassembly or assembly requirements after this test.

TABLE III. Torque requirements.

Fitting Size		03	04	05	06	08	10	12	14	16	20	24
Minimum Torque	in-lb	97	151	173	270	432	594	756	918	1242	1512	1620
	N-m	11	17	20	31	49	67	85	104	140	171	183
Maximum Torque	in-lb	108	168	192	300	480	660	840	1020	1380	1680	1800
	N-m	12	19	22	34	54	75	95	115	156	190	203

3.5.3 Thermal shock. The fitting assembly shall withstand the temperature and pressure when tested in accordance with 4.6.4 without leakage, evidence of permanent deformation, or other malfunction that shall affect assembly or disassembly of the fitting.

3.5.4 Impulse. The fitting assembly shall withstand 200,000 impulse pressure cycles without leakage (see 6.10.7) from the fitting or the fitting-tube junction when tested in accordance with 4.6.5.

3.5.5 Flexure. The fitting assembly shall not leak from the sealing surface or the fitting tube junction prior to completion of ten million cycles at the test stress level specified when tested in accordance with 4.6.6.

3.5.6 Repeated connection. The fitting assembly shall withstand 25 repeated connections at the minimum and maximum torque values specified in Table III when tested in accordance with 4.6.7 without:

a. Leakage at any of the pressure tests of 4.6.7.
b. Inability to assemble the fitting to the point of interface by hand.
c. Nut deformation preventing engagement of nut hex with open end wrench.

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3.5.7 Pneumatic leakage. The fitting assembly shall withstand pneumatic pressure equal to the maximum operating pressure for five minutes at room temperature without any visible bubbles starting after one minute at pressure or other malfunction that would affect assembly or disassembly using the minimum torque values specified in Table III when tested in accordance with 4.6.8.

3.5.8 Fire. The fitting assembly shall withstand a 2000°F (1093°C), 4500 btu/hr (1320 watts) flame for 15 minutes when tested in accordance with 4.6.9. Heat input shall be measured by a .500 inch (12.7 mm) copper tube per SAE-AIR1377. There shall be no leakage detected by visual observation or failure of the test assembly prior to the specified time.

3.6 Part numbering of interchangeable parts. All parts having the same design activity CAGE CODE (Contracting and Government Entity) and part number shall be interchangeable as defined in MIL-STD-280. The item identification and part number requirements of MIL-STD-100 shall govern the design activity part numbers and changes thereto.

3.7 Dimensions and tolerances. All pertinent dimensions, threads and tolerances, where interchangeability, operation, or performance of the fitting may be affected, shall be as specified on the MS drawings or specification sheets.

3.8 Weight. Fitting weights shall not exceed to those values shown on the applicable specification sheets.

3.9 Identification of product. All fittings shall be identified in accordance with 3.9.1 and 3.9.2.

3.9.1 Marking method. Marking shall be impressed, embossed, laser etched, or electrochemical etched in a location and in a manner not detrimental to the fitting. Where practical, markings shall remain visible, after fitting is installed.

3.9.2 Markings. All end fittings and fitting bodies shall be marked with the manufacturer's name or trademark, the material identification code, and complete part number as detailed in the applicable specification sheet. If space permits, the manufacturer's part number and the manufacturer's CAGE CODE shall be marked to ensure positive identification. The manufacturer's part number and drawing number shall be the same.

3.10 Workmanship. Remove all burrs and break all sharp edges. Sealing surfaces shall be free of detrimental, radial, and spiral tool marks. Unless a finer finish is specified on applicable drawings, sealing surfaces shall be smooth to a finish of 45 microinches Ra, as defined in ASME-B46.1. All other machined surfaces shall be smooth to 125 microinches Ra. Unmachined surfaces, such as forging surfaces and bar stock flats shall be free of cracks, laps, and seams.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material (see 6.3).

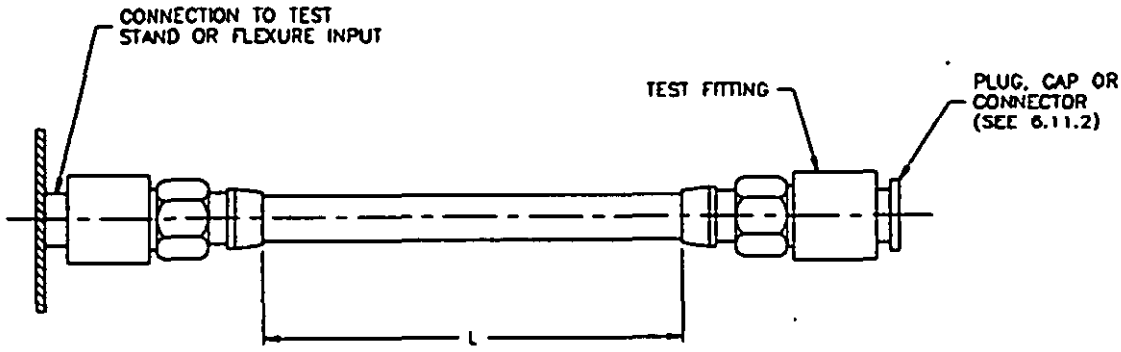
4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

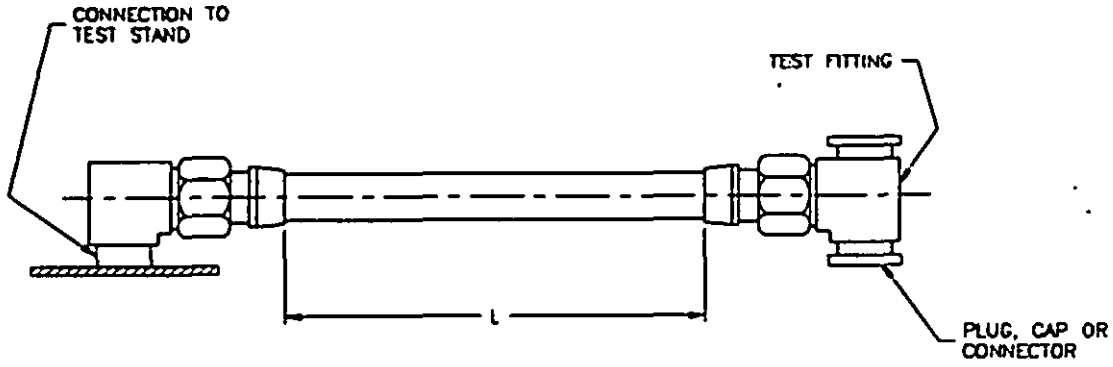
4.3 Qualification inspection. The qualification inspection shall consist of the inspection methods conducted in the order shown in Table IV for the applicable assembly styles (see figure 1).

4.3.1 Test sample for the qualifying activity. The contractor shall submit to the qualifying activity sixteen untested sample fitting assemblies for each size and tubing type in accordance with Table IV (see 6.3). Threaded fittings designed for permanent attachment to adjacent tubing shall be assembled in accordance with the joint design feature (swage, braze or weld). The inspection shall be in accordance with 4.6.1.

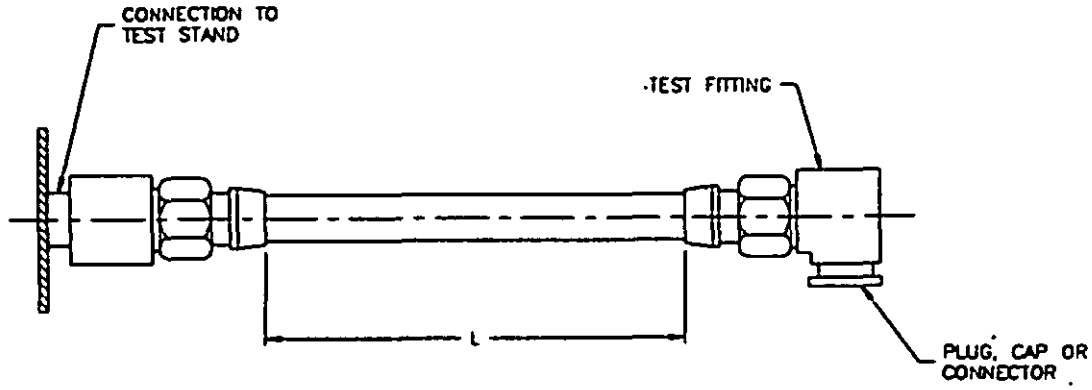
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ASSEMBLY STYLE A



ASSEMBLY STYLE B



ASSEMBLY STYLE C

FIGURE 1. Fitting assembly styles for testing.

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TABLE IV. Fitting assemblies.

Assembly Numbers	* 1-4	5-6	7-12	13-14	15-16
Assembly Style (See Figure 1)	C	A	A	A	B
Length (See Figure 1)	9"	9"	Per SAE-ARP1185	9"	6"
Qualification Test (Perform in Sequence)	4.6.1 4.6.2 4.6.7 4.6.5	4.6.1 4.6.8 4.6.2 4.6.4 4.6.3	4.6.1 4.6.2 4.6.6	4.6.1 4.6.2 4.6.9	4.6.1 4.6.2 4.6.3

*Two samples with 90° elbows and two samples with bulkhead tees.

4.3.2 Retention of qualification. The retention of qualification shall consist of certification by the manufacturer to demonstrate compliance of the qualified fittings with the requirements of this specification. Certification shall be signed by a responsible official of management, attesting that the listed product(s) is still available from the listed plant, can be produced under the same conditions as originally qualified, i.e., same process, materials, construction, design, manufacturer's part number or designations, and meets the requirements of the current issue of the specification. Certification shall be conducted at intervals not exceeding two years. The activity responsible for certification approval is the qualifying activity (see 6.6).

4.4 Quality conformance inspection. The following tests shall be conducted from each lot of fittings submitted for inspection. The test sequence shall be in accordance with Table V (see 6.3).

a. Non-destructive test (see 4.4.2)
b. Destructive test (see 4.4.3)

4.4.1 Inspection sampling. Unless otherwise specified in the contract or purchase order, sample fittings from each lot shall be selected at random in accordance with MIL-STD-105. For non-destructive test the sample inspection shall be level II at an Acceptable Quality Level (AQL), as specified in the contract. For destructive test, the sampling inspection shall be level S-1 at an acceptable quality level of zero percent defects. The sample fittings selected for the destructive test shall be those used for the non-destructive test.

4.4.2 Non-destructive test. The non-destructive test shall be conducted in accordance with the examinations of 4.6.1 and 4.4.2.1.

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TABLE V. Quality conformance test

Test	Design Requirement Paragraph	Inspection Method Paragraph
Non-destructive test Examination of product	3.4.1	4.4.2
Destructive test (perform in sequence) (1) Proof pressure (2) Pneumatic leakage	 3.5.1 3.5.7	 4.6.2 4.6.8

4.4.2.1 Preparation for delivery. All end fittings, fitting bodies, and fitting assemblies shall be visually examined to ascertain that their preparation for delivery shall be in accordance with section 5 of this specification.

4.4.3 Destructive tests. The following destructive tests shall be conducted in sequence in the order specified:

a. Proof Pressure Test 4.6.2.
b. Pneumatic Test 4.6.8.

After the destructive testing, the test samples shall not be shipped or considered part of the contract or order.

4.4.4 Failure of sampling test. When fittings fail to pass a sampling test, the entire lot (see 6.10.8) represented by the sample shall be rejected. The reason for the failures shall be reported (see 6.3).

4.5 Test conditions.

4.5.1 Test fluids. The test fluids used shall be MIL-H-83282 or MIL-H-5606.

4.5.2 Test temperature. All tests except the burst pressure test, the thermal shock test, the impulse test, and the fire test shall be conducted at an ambient temperature of 60°F (16°C) to 100°F (38°C) and a fluid temperature of 60°F (16°C) to 170°F (77°C).

4.6 Inspection methods.

4.6.1 Examination of product. All end fittings, fitting bodies, and fitting assemblies from each lot shall be examined to determine conformance with this specification and the applicable standards with respect to material, design, dimensions, finish, marking, and workmanship.

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4.6.1.1 Material inspection. Fitting and tubing material shall meet the requirements of 3.3. The chemical analysis, mechanical properties, and heat treatment shall be in accordance with the material specifications shown in Table I.

4.6.2 Proof pressure test. Fitting assemblies shall be connected to a pressure source with one end free to move and proof pressure tested at a pressure value equal to two times the nominal operating pressure for a minimum period of five minutes. Use minimum torque values for both ends. Rate of pressure rise shall be 20,000 ±5000 psi (140,000 ±35,000 kPa) per minute. The exterior of the test fitting assembly including the fitting tube joint shall be clean and dry and show no evidence of test fluid prior to testing. At the end of the test, wipe the exterior of fitting assembly with a clean, dry towel and examine for evidence of leakage. Any evidence of fluid on a clean dry towel, after wiping the test connection, shall constitute failure of the test.

4.6.3 Burst pressure test. Fitting assemblies shall be connected to a pressure source and burst pressure tested at a pressure value equal to four times the nominal operating pressure. Use minimum torque values for both ends. The pressure shall be increased at a rate of 20,000 ±5000 psi (140,000 ±35,000 kPa) per minute until the assembly bursts, leaks or four times the operating pressure is obtained. Test samples 5 and 6 shall be tested at the ambient temperature (see 4.5.2), and test samples 15 and 16 shall be tested at the maximum rated temperature of 275°F (135°C) ±10%. No more than one drop of fluid at the joint (both the beam seal end and tube attachment end) is permissible.

4.6.4 Thermal shock test. Use MIL-H-5606 hydraulic fluid for this test. The fitting assembly shall be mounted in a high temperature test set up with one end free to move. A typical test set up is shown in figure 2. After the test fitting is filled with hydraulic fluid, the ambient temperature of the test chamber shall be reduced to -65°F ±2°F (-54°C ±1°C) for a minimum of two hours. At the end of this period, while the test chamber is still at -65°F (-54°C), test fluid at 275°F ±5°F (135°C ±3°C) shall be suddenly introduced into the test assembly at a minimum pressure of 50 psi (350 kPa). Within 15 seconds after the hot fluid has filled the fitting assembly, a proof pressure test shall be performed in accordance with 4.6.2.

4.6.5 Impulse test. Test fitting assemblies shall be tested in accordance with SAE-ARP603 with the following deviations or additions. Fittings for 500 psi (3500 kPa) suction systems as designated in Table II shall not be impulse tested.

a. Duration of test: 200,000 cycles minimum.
b. Temperature: 275°F (135°C) ±10% for 200,000 cycles.
c. Rate of rise: 175,000 to 300,000 psi/second (1,225,000 to 2,100,000 kPa/second).
d. Cycling rate: 70 ±5 cycles per minute.
e. Wave form: Per SAE-ARP603.

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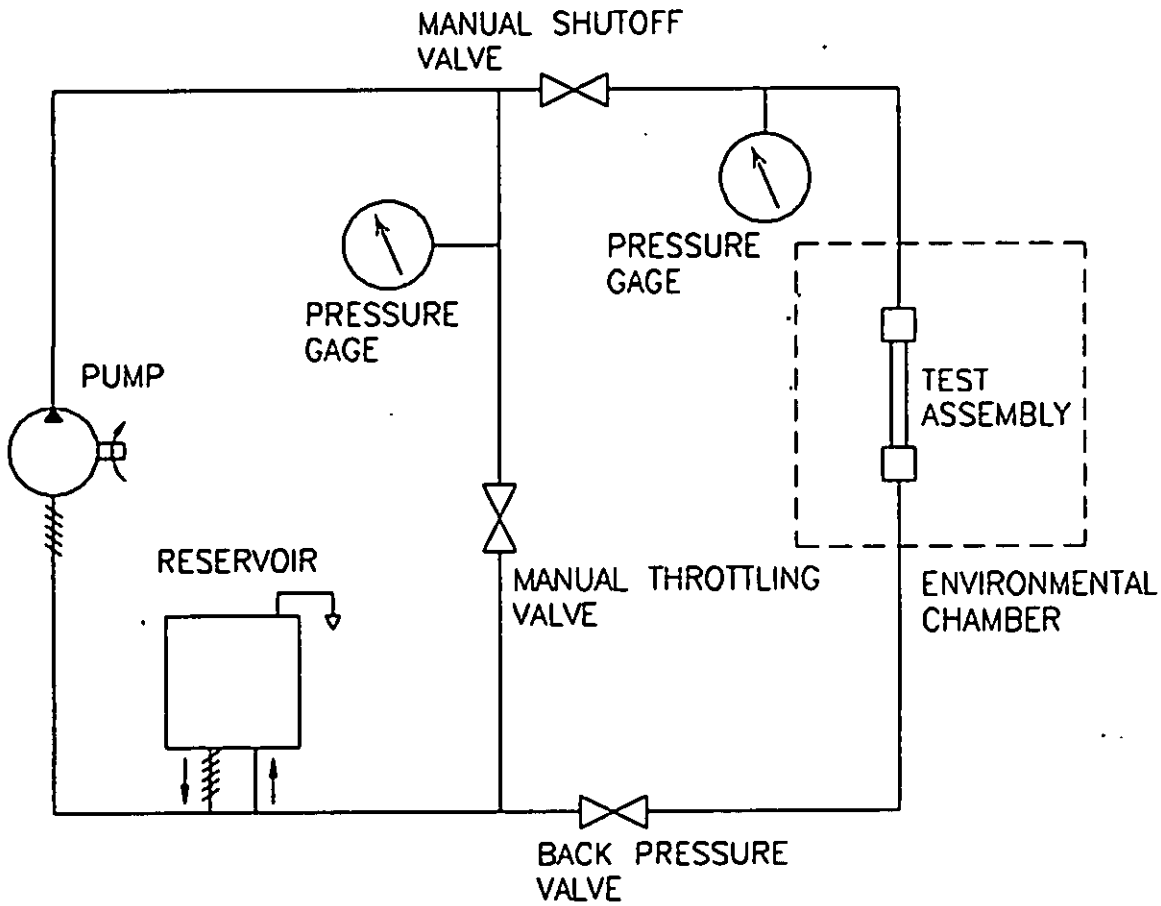


FIGURE 2. Typical setup for thermal shock testing.

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f. Surge peaks: 140 to 157% of nominal operating pressure.
g. Assembly torque: Minimum torque value per Table III.

4.6.6 Flexure test. Fitting assemblies shall be tested in accordance with the flexure test procedure of SAE-ARP1185 except as noted herein. For -03 size, the test assembly shall be five inches (127 mm) in length, test samples shall be assembled using maximum torque values per Table III. Fitting assemblies shall be tested to a minimum bending stress indicated in Table II at room temperature with static internal pressure equal to nominal operating pressure maintained. The bending stress shall be calculated from strain values measured with suitable strain gages on each assembly.

4.6.6.1 Combined stress. The fitting assemblies shall meet the requirements of combined stress for a specific material used. The combined stress shall be the bending stress produced when tested with 4.6.6 plus the axial stress caused by the internal pressure which is calculated by the following formula:

$$S = P\left(\frac{d^2}{D^2 - d^2}\right)$$

where: S = axial tensile stress due to internal pressure
P = internal pressure (operating pressure)
d = tube inside diameter
D = tube outside diameter

4.6.7 Repeated connection. The fitting assemblies shall be tested by connecting and disconnecting each end a total of 25 times. Initially, one end shall be tightened to maximum torque and the other end shall be tightened to minimum torque with no external lubricant added. After each tightening, the fitting assembly shall be disassembled and the sealing faces shall be rotated 60 to 90 degrees with respect to each other. Then the sealing surfaces and threads shall be lubricated with MIL-H-83282 or MIL-H-5606 fluid and the fitting ends shall be reconnected. After the fifth tightening operation, the fitting assembly shall be subjected to the pneumatic test in accordance with 4.6.8 except the maximum and minimum torques shall be retained. For the next tightening operation, the torque at each end shall be switched from minimum to maximum and vice versa. The pneumatic test shall be performed and the torques shall be alternated after every fifth tightening operation. Any leak during the pneumatic test shall constitute a failure of the test.

4.6.8 Pneumatic test. The test assemblies shall be pressure tested at room temperature at a value equal to the nominal operating pressure for a minimum period of five minutes. The test fluid shall be dry, oil free compressed air or nitrogen. The test assemblies shall be prepared without use of oil during assembly or prior to testing. Any oil or grease used for tubing attachment shall be completely removed before test. Test assemblies shall be prepared using minimum torque values specified in Table III. Any visible bubbles or other malfunction that would affect assembly or disassembly shall constitute a failure of the test.

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4.6.9 Fire test. The assemblies shall be mounted in a horizontal plane that lies on the axis of the burner. One end of each test assembly shall be torqued to minimum torque and the other end torqued to maximum torque per Table III. The portion of the flame in which the assembly is to be tested shall deliver a minimum heat output of 4500 btu per hour (1320 watts) as measured by a .500 inch (12.7 mm) copper tube mounted per SAE-AIR1377. Width of the flame shall be such that the 2000°F (1093°C) temperature shall occur (as measured by thermocouple) for a minimum distance of 8 inches (203 mm). The entire fitting connection plus a length of tubing shall be subjected to the flame. During the test, the end connection subjected to the flame shall be subjected to in-line vibration at the rate of 2000 cycles per minute through a total amplitude of not less than .125 inch (3.2 mm), i.e., a displacement of .063 inch (1.6 mm) each side of the neutral position. The assembly shall be subjected to SAE No. 20 oil at a pressure equal to the rated system pressure and at an oil temperature of 200°F-230°F (93°C-110°C). The flow rate shall be flow in gallons per minute equal to 1 x the ID^2 measured in inches. For tolerances and details of test, refer to SAE-AS1055 and SAE-AIR1377.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. All parts shall be free from grease, oil, dirt or any other foreign material. Any process or combination of processes that will accomplish thorough cleaning without damage to the parts shall be acceptable.

5.1.1.2 Preservative. Preservative shall not be used.

5.1.1.3 Unit packing. The unit pack shall be preserved in accordance with Method 1C-10 or Method 1C-3 as specified in MIL-P-116. The threaded ends of the fittings shall be protected with aluminum alloy or rigid plastic caps in accordance with MIL-C-5501. The surface area of the backing board to be used in Method 1C-10 shall be 15 by 12 inches (381 by 305mm). The backing board shall be preprinted or a label enclosed in the unit pack with identification in accordance with MIL-STD-129. The waterproof bag used in Method 1C-3 shall be preprinted or a label enclosed with identification in accordance with MIL-STD-129. The number of items in a unit pack shall be as specified in the contract or order (see 6.2), if it does not specify in the contract or order, it shall be determined in accordance with MIL-STD-2073-1.

5.1.1.4 Intermediate container. Unless otherwise specified, the unit packs shall be packed in intermediate containers in accordance with MIL-STD-2073-1 (see 6.2).

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5.1.2 Commercial preservation. Commercial preservation shall be in accordance with ASTM-D3951.

5.2 Packing. Packing shall be Level A, Level B or Commercial.

5.2.1 Level A. Unit packs preserved as indicated in 5.1.1.3 and packed in intermediate containers shall be packed in wooden containers conforming to MIL-B-43666 Types I and II or PPP-B-601 overseas type.

5.2.2 Level B. Unit packs preserved as indicated in 5.1.1.3 and packed in intermediate containers shall be packed in fiberboard containers conforming to PPP-B-636 weather-resistant type.

5.2.3 Commercial packing. Unit packs preserved as indicated in 5.1.2 and packed in intermediate containers in accordance with ASTM-D3951 shall be packed in outer containers in accordance with ASTM-D3951.

5.3 Marking. Unit packs, intermediate containers and shipping containers shall be marked in accordance with MIL-STD-129. In addition to the requirements of MIL-STD-129, the unit packs, intermediate containers, and shipping containers shall include the following information:

Manufacturer's name or trademark
Manufacturer's part number or CAGE CODE
Acquiring activity code number (if applicable)
Military standard part number
Fitting material
Date of package

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fittings covered by this specification are intended for use in aircraft hydraulic high pressure fluid systems under the conditions specified in paragraph 1.2.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Part number required (see 1.2 and 3.3).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Fitting material and tubing material.
e. Manufacturer's part number or CAGE CODE.

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f. Sampling instructions for tests and when applicable, additional testing of fitting styles (such as tees, reducers, etc.) not included in the test samples (see 4.3.1).
g. Where the qualification verification test samples should be sent and instructions concerning the submittal of the test reports (see 4.3.1).
h. Applicable levels of preservation, packaging, and packing (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 227.405-70 exempts the requirement for a DD Form 1423. For the department of the Navy, the Contract Data Requirements are shown in SD8706.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.1.1	DI-NDTI-80809A	Tests/Inspection Reports	10.2.7
4.4 4.3.1 4.4.4	DI-NDTI-80809A	Tests/Inspection Reports	Use contractor format

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

(Copies of the DoD Federal Acquisition Regulation Supplement and Forms are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001 and the DODSSP-Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

6.4 Sample test fitting assemblies. Sample test assemblies subjected to qualification inspection shall not be considered, accepted or shipped as part of the contract or order.

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6.5 Deletion of sizes 21 and 25. Previous to this specification, sizes 21 and 25 were procured to individual manufacturer standards: These sizes were designed for use on 1.250 inch (31.75 mm) and 1.500 inch (38.10 mm) tube sizes respectively. These fittings duplicated the function of sizes 20 and 24, which were also for use on 1.250 and 1.500 inches (31.75 and 38.10 mm) tubes, but size 20 and 24 were manufactured in corrosion resistant materials only. To eliminate these fittings having the same function, the old -21 has been designated as -20 and the old -25 has been designated as -24. These new -20 and -24 sizes are now manufactured in titanium material. The original -20 and -24 do not exist as a standard part and must be procured to original manufacturer's drawings. Caution must be exercised when purchasing -20 and -24 size replacement fittings for original equipment because the -20 and -24 sizes in this specification will not mate to this older fitting designs.

6.6 Qualification. With respect to products requiring qualification, awards will be made only for products which are qualified for inclusion in the applicable Qualified Products List. The attention of the contractors is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command (AIR-5303), Department of the Navy, Washington, DC 20360; however, information pertaining to qualification of products such as test reports, test fees, and test samples may be obtained from the Commanding Officer, Naval Air Warfare Center Aircraft Division, Warminster, (code 6011), Warminster, PA 18974.

6.7 Record of maintenance. The supplier should maintain a record of inspections for a minimum period of seven years.

6.8 Material certification. Records of chemical analysis and mechanical property tests showing conformance to the applicable material specifications should be made available upon request by the Government Agency. Material certification, heat treatment, and inspection records should be retained for a minimum period of seven years from the date of purchase. The material should conform to the applicable material specification before releasing for production.

6.9 Rejection and retest. Rejected lots should be resubmitted for retest and acceptance as specified in the contract. Parts subjected to qualification tests and failing to conform to the requirements of these tests should be rejected.

6.10 Definitions.

6.10.1 Boss fitting. Boss fitting as used herein is defined as the fitting that connects to MS33649, SAE-AS1300 ring-locked type boss or equivalent.

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6.10.2 Connector. Connector as used herein is defined as a replaceable component which connects end fittings, bosses or ports by means of threads. Generally these components are end fittings with internal or external threads and fitting bodies.

6.10.3 End fitting. End fitting as used herein is defined as fitting permanently attached to tubing end. An end fitting with an internally threaded nut is defined as a female end fitting. An end fitting with an externally threaded end is defined as a male end fitting.

6.10.4 Failure. Failure as used herein is defined as any deformation that would contribute to seal leakage or would promote premature fatigue breakage, unless otherwise determined to be due to a tubing defect.

6.10.5 Fitting assembly. Fitting assembly as used herein is defined as the end fittings attached to the appropriate tubing with necessary connectors to conform to proper test sample (see 4.3.1 and Table IV).

6.10.6 Fitting body. Fitting body as used herein is defined as a replaceable component which connects male or female end fittings, bosses, or ports by means of threads. Generally these fittings are unions, adapters, 45 and 90 degree elbows, tees, and crosses.

6.10.7 Leakage. Leakage as used herein is defined as the escape of fluid (gaseous or liquid) from any point of the fitting assembly including the fitting-tube junction. This junction is defined to include one tube outside diameter in length beyond the fitting envelope. The envelope is the area from free parent tube material to free parent tube material or boss.

6.10.8 Lot. A lot is defined as parts consist of all parts of a given part number made from the same batch of material and processed at the same time.

6.10.9 Torque. Torque as used herein is defined as the necessary loads (or assembly stresses) to enssure the sealing of the end fitting to the fitting body.

6.11 Subject term (key words) listing.

Connector
Metal-to-metal seal
Operating Temperature -65°F to +275°F (-54°C to +135°C)
Operating pressure 3000/4000 psi (21,000/28,000 kPa)
Tubing connection

6.13 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:
Army - AV
Navy - AS
Air Force - 99

Review activities:
Army - AT, ME, MI
Air Force - 71, 82
DLA - CS

Preparing activity:
Navy - AS
(Project No. 4730-1027)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.
   NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-F-85421B	2. DOCUMENT DATE (YYMMDD) 940829

3. DOCUMENT TITLE
FITTINGS, TUBE, FLUID SYSTEMS, SEPARABLE, BEAM SEAL, 3000/4000 PSI, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial: (2) DSN: (*If Applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE NUMBER (*Include Area Code*) (1) Commercial: (908) 323-7488 (2) DSN: 624-7488
c. ADDRESS (*Include Zip Code*) CODE SR3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	*IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:* Defense Quality and Standardization Office, 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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